

Sample Question Paper

First Year Engineering - Basic Mathematics

1. Define a matrix and give one example.
2. Evaluate: $(2x + 3)(x - 4)$.
3. Find the derivative of $x^2 + 5x + 6$.
4. State Pythagoras' theorem.
5. Find the value of $\sin 30^\circ$.
6. Solve: $3x + 5 = 20$.
7. What is the difference between scalar and vector quantities?
8. Find the area of a circle of radius 7 cm.
9. Write the formula for compound interest.
10. What is the transpose of a matrix?